CURRICULUM VITAE

Steve O'Brien

Engineer · Founder · CEO & CTO



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I build software, AI systems, and technology businesses. Over more than 20 years, I've combined hands-on engineering with product and commercial leadership across healthcare, finance, and aerospace. I founded Newicon and led it through acquisition. Today I build AI products and advise founders and leadership teams on technology and delivery.

EXPERIENCE

Independent 2026 - present

Engineer, Fractional CTO & AI Product Builder

- Building db3.ai, a TypeScript framework for AI applications, with model orchestration, agents, provider failover, and usage and cost tracking.
- Developing DOM Studio, a Vue UI system for AI conversations and agent workflows, and GrowthScout, an AI-assisted search intelligence and publishing product.
- Advising founders and leadership teams on product direction, architecture, delivery, and AI adoption.

Newicon Ltd 2009 - 2026

Founder, CEO & CTO

- Founded and led a Bristol technology company, building products for founders and established organisations across healthcare, finance, and aerospace.
- Built products and platforms primarily with PHP, Node.js, and web technologies.
- Combined CEO and CTO responsibilities with hands-on architecture, engineering, CI/CD, automated testing, and DevOps. Clients included Airbus, Thales, UWE, and Palladium.
- Grew Newicon to £1.5m turnover alongside a £0.5m digital marketing business that became Flex Digital.
- Led Newicon through its 2025 acquisition by Linebreak; continued as CTO through January 2026.

Airbus UK, Filton Sep 2006 - Jun 2009

Systems Engineer, Modelling & Simulation

- Developed modelling, simulation, and test software using C++, MATLAB, and .NET.
- Designed AutoTR and its code interpreter to automate landing-gear testing, from simulation to hardware-in-the-loop rigs. Received an Airbus international Award for Excellence.
- Built LGTest (PHP/MySQL) to give global teams immediate access to controlled test results, replacing manually compiled data sent around the world on DVDs.

Airbus UK, Filton Sep 2002 - Sep 2006

Technical Apprentice

- Completed a four-year Technical Apprenticeship and HND in Aeronautical Engineering. Named Airbus Apprentice of the Year three years in succession.

SELECTED PROJECTS

Internal due diligence AI agent

Built an internal agent for Newicon's acquisition, using embeddings, vector search, retrieval-augmented generation (RAG), and OpenAI function calling to find evidence across company documents and policies.

Application frameworks from first principles

Designed and built my own full-stack frameworks from scratch in PHP and TypeScript, covering data systems, authentication, real-time WebSockets, and operational infrastructure. Applications include regulated financial services products such as the Hartley pension portal.

EarSwitch assistive communication device

Newicon built data capture and OpenCV analysis for Bath University research, then integrated Python software with the in-ear device and a caregiver app. The project secured £1.2m investment and UK Innovative Devices Access Pathway selection.

Airbus Oleo measurement app

Led an early iPhone computer-vision app using OpenCV to measure landing-gear shock absorbers and give engineers an immediate pass/fail result.

GrowthScout search intelligence platform

Leading product and full-stack TypeScript engineering for AI visibility tracking, keyword research, website audits, and content planning through to writing and publishing.

TECHNICAL SKILLS

I have deep technical experience across many languages and stacks. I work from fundamental programming concepts and choose the best tool for the job, understanding when Rust, Go, Node.js, or C best suits the problem, performance requirements, and team.

Languages: JavaScript · TypeScript · PHP · C++ · MATLAB · Python

Frameworks and runtimes: Node.js · .NET · Vue 3 · React · Next.js · Fastify · Laravel

Frontend and real-time: Web Components · WebSockets · SSR · Pinia

Backend and data: PostgreSQL · MariaDB / MySQL · REST APIs · Background workers · Authentication · Stripe

AI APIs and systems: OpenAI API · Anthropic API · OpenRouter API · LLM applications · Open-weight LLMs · Agent systems · Embeddings · Vector search · Knowledge graphs

Engineering tools and delivery: Git · Docker · Linux · CI/CD · DevOps · Automated testing · Vitest · Playwright

RECOGNITION & SPEAKING

- Insider South West 42 under 42, 2022 cohort
- Bristol Technology Festival speaker, including a 2022 talk on the future of IoT
- Podcast guest on technology, leadership, innovation, and company building

INTERESTS

Following the Thousand Brains Project · Robotics · Coding for fun · Playing guitar and piano